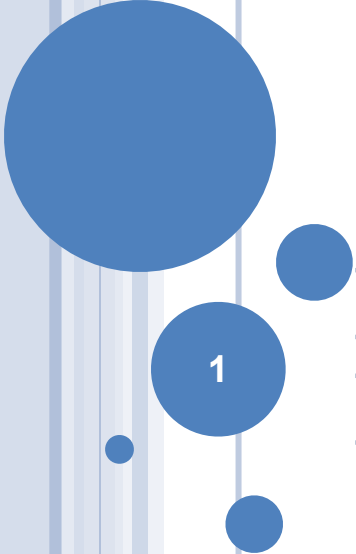


INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION



Chapter 4

Probability and Probability Distributions

1

A decorative vertical bar on the left side of the slide, composed of several thin, parallel lines in various shades of blue. To the right of this bar, there are several overlapping circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, and several smaller circles are arranged below it, creating a cascading effect.

4.1 EVENTS AND SAMPLE SPACE

2

WHAT IS PROBABILITY?

- In Chapters 2 and 3, we used graphs and numerical measures to describe data sets which were usually **samples**.
- We measured “how often” using

$$\text{Relative frequency} = f/n$$

Sample $\longrightarrow$ Population
And “How often”
= Relative frequency $\longrightarrow$ Probability

Ch 1, 2, 3	Ch 4, 5, 6
Sample	Population
Data (measurements of a variable): qualitative, quantitative (discrete, continuous)	Random variable: quantitative (discrete, continuous)
$x_1, \dots, x_n$	X
Relative frequency	Probability
Graphs (histogram, boxplot, dot plot, stemplot)	Probability mass function (pmf), probability density function (pdf)

BASIC CONCEPTS

- An **experiment** is the process by which an observation (or measurement) is obtained.
- **Experiment: Record an age**
- **Experiment: Toss a dice**
- **Experiment: Record an opinion (yes, no)**
- **Experiment: Toss two coins**

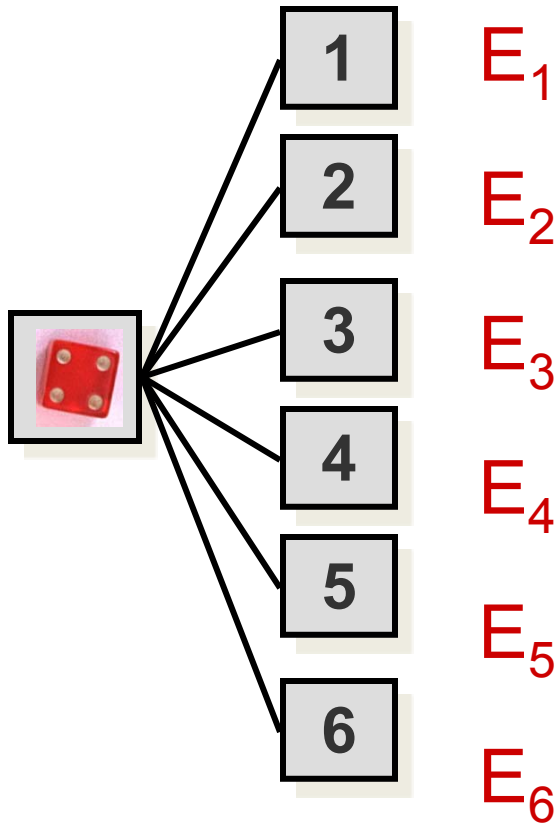
BASIC CONCEPTS



- A **simple event** is the outcome that is observed on a single repetition of the experiment.
 - The basic element to which probability is applied.
 - One and only one simple event can occur when the experiment is performed.
- A **simple event** is denoted by E with a subscript.

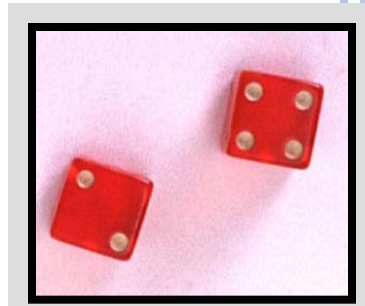
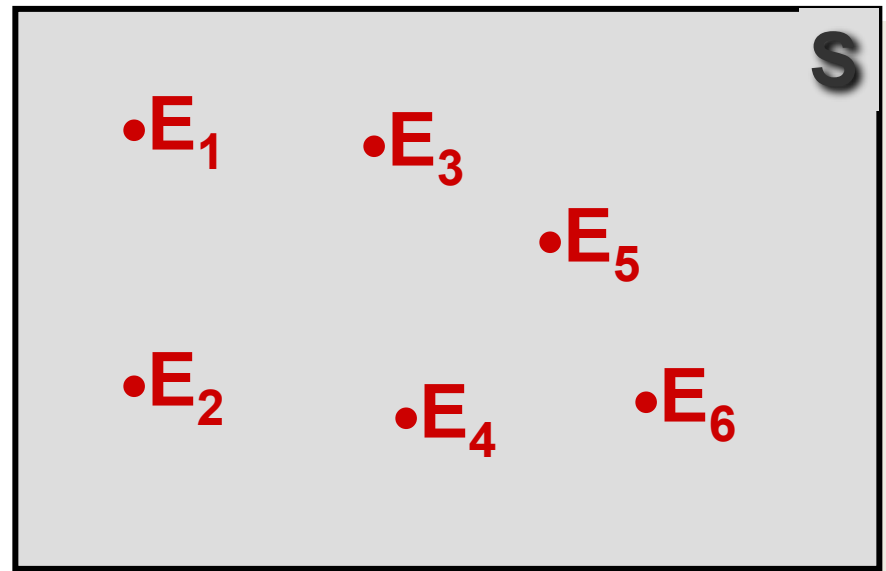
EXAMPLE

- **The die toss:**
- Simple events:



Sample space:

$$S = \{E_1, E_2, E_3, E_4, E_5, E_6\}$$

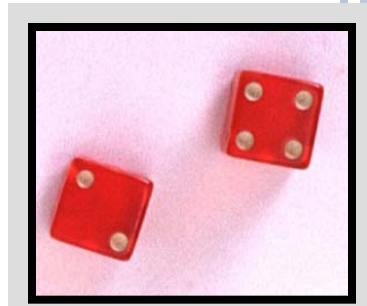


BASIC CONCEPTS



- Each simple event will be assigned a **probability**, measuring “how often” it occurs.
- The set of all simple events of an experiment is called the **sample space, S** .
- The set of no events of an experiment is called the **empty set, ϕ** .

BASIC CONCEPTS



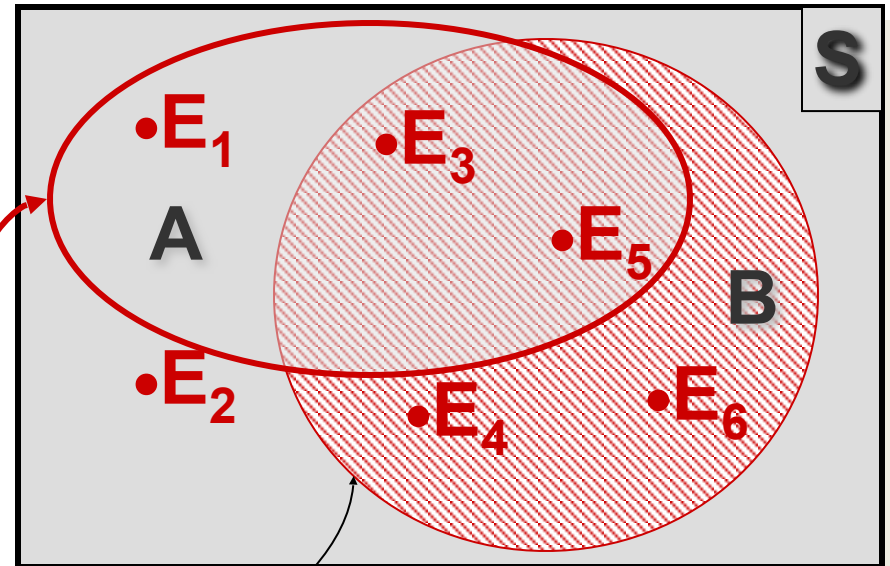
- An **event** is a collection of one or more simple events.

- **The die toss:**

- A: an odd number
- B: a number > 2

$$A = \{E_1, E_3, E_5\}$$

$$B = \{E_3, E_4, E_5, E_6\}$$



BASIC CONCEPTS

- Two events are **mutually exclusive** if, when one event occurs, the other cannot, and vice versa.



•Experiment: Toss a die

–A: observe an odd number

–B: observe a number greater than 2

–C: observe a 6

–D: observe a 3

$$\Omega = \{1,2,3,4,5,6\}$$

$$A = \{1,3,5\}$$

$$B = \{3,4,5,6\}$$

$$C = \{6\}$$

$$D = \{3\}$$

B and C?
C and D?

$B \cap C = \{6\}$. Not
Mutually Exclusive

$C \cap D = \phi$. Mutually Exclusive

Some experiments can be generated in stages, and the sample space can be displayed in a **tree diagram**. Each successive level of branching on the tree corresponds to a step required to generate the final outcome.

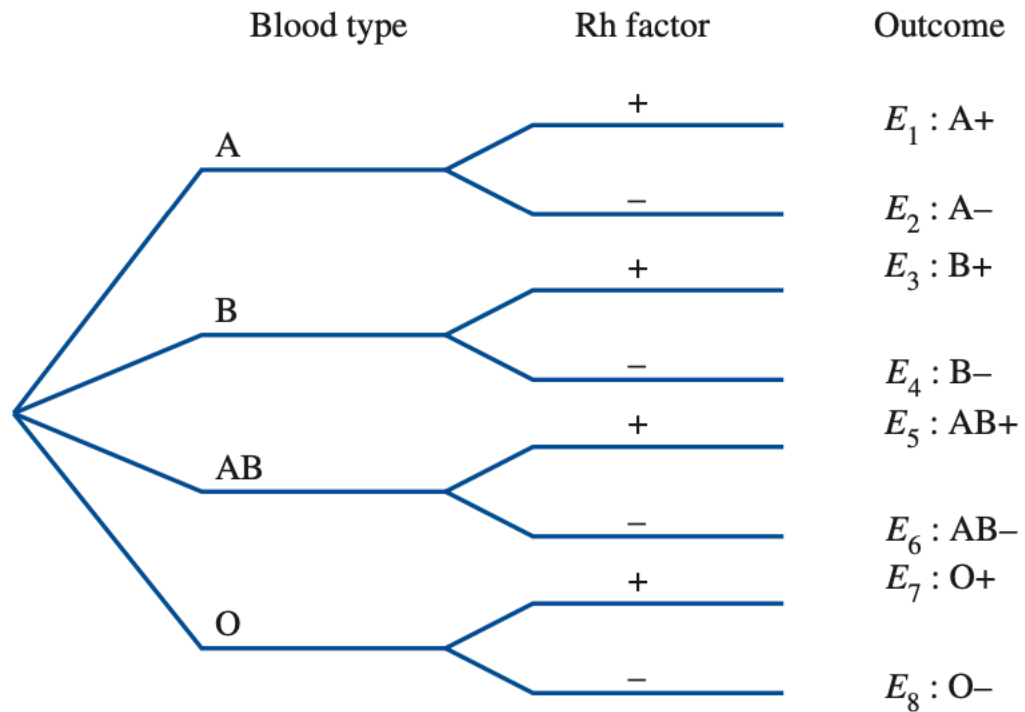
EXAMPLE 4.4

A medical technician records a person's blood type and Rh factor. List the simple events in the experiment.

Solution For each person, a two-stage procedure is needed to record the two variables of interest. The tree diagram is shown in Figure 4.2. The eight simple events in the tree diagram form the sample space, $S = \{A+, A-, B+, B-, AB+, AB-, O+, O-\}$.

Figure 4.2

Tree diagram for
Example 4.4



■ **Table 4.1 Table of Outcomes for Example 4.4**

Rh Factor	Blood Type			
	A	B	AB	O
Negative	A−	B−	AB−	O−
Positive	A+	B+	AB+	O+

The left side of the slide features a series of vertical stripes in various shades of blue. Overlaid on these stripes are several circles of different sizes, also in shades of blue. One circle contains the number 14.

4.2 CALCULATING PROBABILITIES USING SIMPLE EVENTS

14

THE PROBABILITY OF AN EVENT



- The probability of a simple event E measures “how often” we think A will occur. We write $P(E)$.
- Suppose that an experiment is performed n times. The relative frequency for a simple event E is

$$\frac{\text{Number of times } A \text{ occurs}}{n} = \frac{f}{n}$$

- If we let n get infinitely large,

$$P(E) = \lim_{n \rightarrow \infty} \frac{f}{n}$$

Difficult to calculate in practice!

THE PROBABILITY OF AN EVENT: MATH AXIOMS



1. $P(A)$ must be between 0 and 1.

$$0 \leq P(A) \leq 1 \quad \leftarrow \text{Relative frequency!}$$

2. The sum of the probabilities for all simple events in S equals 1.

$$P(S) = 1$$

3. The probability of an event A is found by adding the probabilities of all the simple events contained in A :

$$P(A) = P(E_1, \dots, E_k) = P(E_1) + \dots + P(E_k)$$

S.O.P to find a probability

1. Write down the sample space and simple events:

$$S = \{E_1, E_2, \dots, E_N\}$$

2. Identify the event A using simple events to find the number of simple events in A , n_A

3. Then, $P(A) = \frac{n_A}{N}$

Note

- We denote an empty set by $\{\}$ or ϕ .
- For an empty set, ϕ , $P(\phi) = 0$
- If event A can never occur, $P(A) = 0$.
- If event A always occurs when the experiment is performed, $P(A) = 1$.

COUNTING RULES

Axiom 3: The **probability of an event** A is found by adding the probabilities of all the simple events contained in A .

If the simple events in an experiment are **equally likely**, you can calculate

$$P(A) = \frac{n_A}{N} = \frac{\text{number of simple events in } A}{\text{total number of simple events}}$$

You can use **counting rules** in Ch 4.3 to find n_A and N .

FINDING PROBABILITIES



- Probabilities can be found using
 - Estimates from empirical studies
 - Common sense estimates based on equally likely events

•Examples:

- Toss a fair coin. $P(\text{Head}) = 1/2$
- 10% of the U.S. population has red hair.
Select a person at random.

$$P(\text{Red hair}) = .10$$



EXAMPLE

- Toss a fair coin twice. What is the probability of observing at least one head?

1st Coin	2nd Coin	E_i	$P(E_i)$
H	H	HH	1/4
	T	HT	1/4
T	H	TH	1/4
	T	TT	1/4

$$\begin{aligned} &P(\text{at least 1 head}) \\ &= P(E_1) + P(E_2) + P(E_3) \\ &= 1/4 + 1/4 + 1/4 = 3/4 \end{aligned}$$

EXAMPLE

- A bowl contains three M&Ms[®], one red, one blue and one green. A child selects two M&Ms at random. What is the probability that at least one is red?

1st M&M	2nd M&M	E_i	$P(E_i)$
		RB	1/6
		RG	1/6
		BR	1/6
		BG	1/6
		GB	1/6
		GR	1/6

$$\begin{aligned} P(\text{at least 1 red}) &= P(RB) + P(BR) + \\ &P(RG) + P(GB) \\ &= 4/6 = 2/3 \end{aligned}$$

EXAMPLE 4.6

The proportions of blood types A, B, AB, and O in the population of all Caucasians in the United States are reported as .40, .11, .04, and .45, respectively.¹ If a single Caucasian is chosen randomly from the population, what is the probability that he or she will have either type A or type AB blood?

Solution The four simple events, A, B, AB, and O, do *not* have equally likely probabilities. Their probabilities are found using the relative frequency concept as

$$P(A) = .40 \quad P(B) = .11 \quad P(AB) = .04 \quad P(O) = .45$$

The event of interest consists of two simple events, so

$$\begin{aligned} P(\text{person is either type A or type AB}) &= P(A) + P(AB) \\ &= .40 + .04 = .44 \end{aligned}$$

KEY CONCEPTS

I. Experiments and the Sample Space

1. Experiments, events, mutually exclusive events, simple events
2. The sample space
3. Venn diagrams, tree diagrams, probability tables

II. Probabilities

1. Relative frequency definition of probability
2. Properties of probabilities
 - a. Each probability lies between 0 and 1.
 - b. Sum of all simple-event probabilities equals 1.
3. $P(A)$, the sum of the probabilities for all simple events in A

KEY CONCEPTS

III. Counting Rules

1. mn Rule; extended mn Rule

2. Permutations:

$$P_r^n = \frac{n!}{(n-r)!}$$

3. Combinations:

$$C_r^n = \frac{n!}{r!(n-r)!}$$

IV. Event Relations

1. Unions and intersections

2. Events

a. Disjoint or mutually exclusive: $P(A \cap B) = 0$

b. Complementary: $P(A) = 1 - P(A^C)$

KEY CONCEPTS

3. Conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

4. Independent and dependent events

5. Additive Rule of Probability:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

6. Multiplicative Rule of Probability:

$$P(A \cap B) = P(A)P(B | A)$$

7. Law of Total Probability

8. Bayes' Rule

KEY CONCEPTS

V. Discrete Random Variables and Probability Distributions

1. Random variables, discrete and continuous
2. Properties of probability distributions

$$0 \leq p(x) \leq 1 \text{ and } \sum p(x) = 1$$

3. Mean or expected value of a discrete random variable: Mean : $\mu = \sum xp(x)$

4. Variance and standard deviation of a discrete random variable:
Variance : $\sigma^2 = \sum (x - \mu)^2 p(x)$

$$\text{Standard deviation : } \sigma = \sqrt{\sigma^2}$$